AI PRACTICE SCORED REPLAY

Design a Metrics Pipeline - Scored Mock Replay

A 38-minute interview-shaped replay covering the data contract, workload, architecture, correctness, failure behavior, transcript evidence, score reason, and precise correction.

Define the user-visible freshness and completeness contract before topology, then make delivery semantics and failure behavior audible at the moment they change the design.

VERSION 1.0

FORMAT SCORED MOCK REPLAY + CORRECTION LAB

STATUS LEARNER READY

AI-PRACTICE / METRICS-PIPELINE-SCORED-REPLAY

START FROM THE USER-VISIBLE CONTRACT

Design a metrics pipeline by deciding what must remain true

A strong answer does not begin with a queue or a database. It defines who emits data, what acknowledgement means, how fresh results must be, which queries matter, and how loss, duplication, lateness, cardinality, and cost will be bounded.

TEACHING-DATA BOUNDARY

This is a scripted teaching replay. Names, timestamps, workloads, scores, incidents, and outcomes are fictional sample data and do not represent a learner or employer. Adapt the method and use only verified evidence.

01	02	03	04	05
Promise	**Semantics**	**Flow**	**Pressure**	**Evidence**
State freshness, completeness, query latency, retention, and isolation targets.	Define the sample, series identity, timestamps, acknowledgement, and replay rules.	Separate admission, durable buffering, processing, storage, and query serving.	Test cardinality, skew, late data, hot series, store failure, and regional loss.	Close with measurable SLIs, accepted costs, and a reversal condition.

The 38-minute interview contract

Clock	Decision to make	Evidence to leave audible
0-5	Frame users, alerts, dashboards, scale, freshness, retention, and non-goals.	A one-sentence contract and the first uncertainty to resolve.
5-9	Estimate samples, bytes, series, write rate, query rate, and retention pressure.	Rounded ranges connected to architecture decisions, not false precision.
9-16	Define metric types, label policy, event time, ingest time, and acknowledgement.	A stable sample envelope and explicit delivery semantics.
16-24	Choose partitioning, durable buffer, processors, hot storage, rollups, and cold retention.	Every component tied to a requirement or failure boundary.
24-31	Walk query, late-data, replay, overload, and tenant-isolation paths.	User-visible behavior during partial failure.
31-38	Compare trade-offs, operating signals, alternatives, and the close.	Accepted cost, trigger to revisit, and the highest-risk assumption.

Clarify before drawing

- [ ] Who emits samples: services, hosts, devices, customer workloads, or internal agents?
- [ ] Which product actions depend on the pipeline: alerts, dashboards, investigations, billing, or compliance?
- [ ] What does accepted mean, and may an acknowledged sample later be missing from a query?
- [ ] What freshness and query-latency targets matter by use case and time window?
- [ ] Which labels are permitted, and which identifiers or sensitive values must never become dimensions?
- [ ] How long must raw and rolled-up data remain available, and which historical questions need full resolution?

MAKE THE DATA CONTRACT EXPLICIT

Metric semantics determine whether the answer is correct

Storage can be replaced. Ambiguous samples, unbounded labels, and undefined late-data behavior leak into every downstream component. State these rules before choosing the store.

Concept	Meaning	Interview consequence
Counter	A cumulative value that normally increases until a reset.	Compute change over a time window and detect resets; do not treat the absolute value as a rate.
Gauge	A point-in-time value that may rise or fall.	Define sampling and staleness behavior; last value is not automatically current truth.
Histogram	Counts observations in compatible value buckets.	Aggregate bucket counts before estimating quantiles; never average percentiles.
Series identity	Metric name plus the complete normalized label set.	Cardinality is unique series count, not simply the number of label keys.
Event time	When the measurement occurred at the source.	Needed for correct windows when transport or processing is delayed.
Ingest time	When the platform accepted the sample.	Useful for freshness and operational lag, but not a substitute for event time.

FICTIONAL WORKED EXAMPLE - ADAPT THE METHOD, NOT THE FACTS

Rounded assumption	Pressure created	Decision it informs
500,000 samples/s	The admission and log path must sustain burst headroom above the average.	Partition writes and enforce per-tenant admission rather than relying on one shared endpoint.
40-80 encoded bytes/sample	About 1.7-3.5 TB/day before indexes and replication.	Compress, tier retention, and keep full-resolution duration explicit.
20 million active series	Catalog memory, compaction, and query fan-out can dominate raw byte cost.	Set label budgets, reject unsafe dimensions, and isolate tenant explosions.
Alerts need under 60 s freshness	Long batch intervals or an opaque replay can violate the primary promise.	Track end-to-end freshness and reserve a low-latency processing path.

SIZING RULE

Use a range and show the unit chain: samples/second x bytes/sample x 86,400 seconds/day. Then name what the estimate excludes, such as indexes, replicas, metadata, compaction, and headroom.

KEEP INGESTION AND QUERIES INDEPENDENTLY SURVIVABLE

Name each boundary, source of truth, and replay path

The architecture is valuable only when the listener can trace a sample from emission to a visible query result and explain what happens if any stage slows or fails.

01	02	03	04	05
Collect	**Admit**	**Buffer**	**Process**	**Serve**
Batch locally, apply backpressure, and attach stable source identity.	Authenticate tenant, validate schema, enforce label and rate budgets.	Acknowledge at the chosen durable boundary and partition for bounded skew.	Normalize, deduplicate within scope, aggregate, and manage late events.	Use hot time-series storage plus rollups, cold data, and a bounded query plan.

Stage	Contract	Failure question	Evidence
Agent/SDK	Local buffering has a byte/time limit.	What is dropped or blocked when the device is offline?	Dropped-sample count and oldest buffered age.
Admission	Only valid, authorized, bounded-cardinality samples enter.	Can one tenant exhaust shared capacity?	Rejected samples by reason and tenant budget saturation.
Durable log	Acknowledged data survives a process restart within the stated durability target.	What ordering and replay guarantees exist per partition?	Oldest unprocessed age, partition skew, and under-replicated state.
Processor	Retries cannot multiply effects or move the watermark silently.	How are duplicates, late data, and counter resets handled?	Replay duplicates, correction count, checkpoint age.
Query tier	Fresh and historical results expose known staleness or approximation.	What happens when hot storage or a region is unavailable?	Query latency, partial-result flag, freshness age, error budget.

BOUNDARY

Write path owns acceptance

Declare the exact acknowledgement boundary. A durable log may be the replay source, but only if retention and recovery meet the accepted-sample contract.

BOUNDARY

Read path owns interpretation

A query must reveal staleness, partial coverage, rollup resolution, or approximation. A fast number without its meaning is not a reliable result.

PRESSURE-TEST CORRECTNESS BEFORE OPTIMIZING COST

Design the failure behavior a learner can defend

At-least-once delivery is useful only when duplicate scope, idempotency, late data, replay, and query-visible correction are explicit. Exactly-once claims must be limited to a named boundary.

Pressure	Unsafe shortcut	Defensible policy
Duplicate replay	Assume the queue prevents every duplicate.	Use a stable source/sample identity, a bounded deduplication horizon, and idempotent aggregate updates or reconciliation.
Late or out-of-order sample	Drop it without telling query users.	Define watermark and lateness allowance; mark provisional windows and publish corrections when policy permits.
Counter reset	Interpret a negative delta as negative usage.	Detect restart/reset markers, split the interval, and expose unknown data when the reset cannot be reconstructed.
Hot series or tenant	Add partitions without a fairness rule.	Enforce tenant/series budgets, isolate heavy workloads, and choose a partition key with a stated skew trade-off.
Store outage	Keep accepting forever with no capacity bound.	Use a durable buffer with time/byte limits, shed by explicit priority, and communicate freshness degradation.
Wide historical query	Fan out without a cost or deadline budget.	Select rollup resolution, bound fan-out, cache safe results, and return partial/timeout semantics deliberately.

Pipeline health must be observable from the user's promise

+ End-to-end freshness from event time to query visibility, split by tenant and priority.
+ Accepted-sample completeness, explicit drop/reject reasons, and correction/replay volume.
+ Oldest unprocessed age and partition skew, not only average consumer lag.
+ Query latency and error/partial-result rate by time range and resolution.
+ Active-series cardinality, new-series creation rate, hot-series concentration, and budget rejections.
+ An alert includes an owner and action; dashboards alone do not create recovery.

TRUST BOUNDARY

Never place secrets, raw personal identifiers, request IDs, or unbounded user-controlled values in metric labels. Sampling is not automatically safe for billing, security, or other authoritative events.

INSPECT THE ANSWER AT THE DECISION MOMENT

Annotated scored replay: preserve what worked, repair what did not

A replay is not a rewritten perfect answer. It quotes the observable move, keeps the correct reasoning, identifies the missing decision, and adds one sentence that changes the downstream design.

FICTIONAL WORKED EXAMPLE - ADAPT THE METHOD, NOT THE FACTS

Moment	What the candidate did	Evidence and gap	Precise improved sentence
06:14	Separated ingestion from query serving and gave each an independent scaling path.	Correct boundary. The acknowledgement point and accepted-sample durability were still unstated.	I will acknowledge after the tenant-validated sample is durably appended; query freshness can degrade without losing that accepted record.
14:52	Added replication for the time-series store.	Useful availability instinct. The answer did not choose stale, partial, or unavailable query behavior during failover.	During failover I will serve the last complete window with its freshness age; I will not merge an incomplete replica as if it were current.
27:08	Suggested adding workers while monitoring queue length.	Capacity may help, but depth alone hides oldest-event SLO breach, partition skew, retry amplification, and downstream headroom.	I will scale only after confirming parallel partitions and downstream capacity, while tracking oldest unprocessed age and successful throughput.

Follow-up probes that reveal real understanding

Probe	Strong decision to expose
Can you guarantee exactly once?	Scope the guarantee, name duplicate boundaries, and explain idempotent effects plus reconciliation.
A customer adds user_id as a label. What happens?	Reject or transform it under a documented cardinality/privacy policy and show the rejection signal.
How do you aggregate p99 latency?	Aggregate compatible histogram buckets or a mergeable sketch; do not average precomputed percentiles.
A sample arrives ten minutes late. Is history rewritten?	Apply the declared watermark/correction policy and make provisional or corrected results visible.
Why not store raw data forever?	Tie retention and rollups to query value, cost, privacy, and the questions that still need full resolution.

EXPLAIN THE REPRESENTATIVE SCORE

Every point must trace back to evidence

The catalog's scripted score is a teaching example, not learner history or a readiness verdict. The dimensions below average to 78 only because the same prompt, clock, and evidence record are used.

FICTIONAL SCRIPTED SCORECARD - ADAPT THE RUBRIC, NOT THE SCORE

Dimension	Score	Evidence	Highest-value next move
Problem framing	82	Users and ingest/query separation were clear by 06:14.	State freshness, completeness, retention, and acknowledgement in the opening contract.
Technical depth	76	Replication and scaling mechanisms appeared, but semantics and replay were incomplete.	Choose delivery, deduplication, late-data, and counter-reset behavior before topology detail.
Trade-off clarity	70	Alternatives were named, but accepted costs and reversal triggers were rarely explicit.	For each major choice, name the cost, user-visible failure behavior, and trigger to revisit.
Communication	84	The flow was traceable and interviewer probes were answered directly.	Front-load the decision sentence so strong detail does not arrive before its purpose.

SCORE REASON

Why 78, not 90

The answer was coherent and organized, but key correctness semantics arrived late or remained implied. A high score requires those choices before storage topology and under failure probes.

EVIDENCE RULE

What not to infer

Do not compare this number with a different prompt, interviewer, rubric, or assistance condition and call the difference improvement. Normalize conditions before claiming movement.

ONE CORRECTION

Before naming a store, say: accepted samples are durably buffered with at-least-once delivery; duplicate effects are bounded by stable identity; late samples follow a visible correction policy; alerts optimize freshness while historical queries may use rollups.

Comparable re-test exit criteria

- [ ] Use the same prompt family, 38-minute clock, rubric, and assistance rules.
- [ ] Semantics appear before topology without an interviewer reminder.
- [ ] The late-data, replay, failover, and hot-cardinality probes each receive a user-visible policy.
- [ ] A chosen trade-off includes accepted cost and a measurable reversal trigger.

RUN THE REPLAY AS DELIBERATE PRACTICE

Turn one transcript into a stronger adjacent answer

The goal is transfer: the learner should make the corrected decision in a nearby telemetry or event-stream prompt without copying a memorized architecture.

A stronger 90-second opening

I will design for multi-tenant infrastructure metrics used by alerts and dashboards. Before topology, I want to confirm sample rate, active-series cardinality, alert freshness, query windows, retention, and accepted loss. My initial contract is that a tenant-valid sample is acknowledged only after durable append, then processed at least once with bounded duplicate effects. Event time drives windows; late samples follow a stated correction policy. I will separate the write path from query serving so a query-tier failure can degrade freshness without losing accepted samples. Next I will size the write and series pressure, then choose partitioning, hot retention, rollups, and the failure behavior for overload and replay.

Practice block	Action	Proof of completion
Replay - 8 min	Listen/read only around 06:14, 14:52, and 27:08; record the decision that was audible and the missing consequence.	Three evidence notes with timestamps; no general impression words.
Repair - 8 min	Say the improved opening and one corrected sentence for each moment without adding invented requirements.	Semantics precede store choice; each correction changes a downstream decision.
Pressure - 10 min	Answer duplicate replay, high-cardinality label, late event, hot tenant, and regional failover probes.	Each response contains policy, signal, and user-visible behavior.
Transfer - 9 min	Use a nearby clickstream or device-telemetry prompt with different scale and query needs.	The method transfers while the architecture changes with the contract.
Close - 3 min	Choose one remaining weakness and schedule a comparable re-test.	One behavior, one evidence source, one stop condition.

Completion checklist

- [ ] The answer states user-visible freshness, completeness, latency, retention, and isolation before components.
- [ ] Sample schema, timestamps, series identity, label budget, acknowledgement, delivery, and replay are explicit.
- [ ] Capacity math uses rounded ranges and connects to partitioning, compression, retention, or rollups.
- [ ] Every failure probe ends with observable behavior, not only a mechanism name.
- [ ] The corrected response preserves what was already correct and adds only the missing decisions.
- [ ] The final re-test uses comparable conditions and records evidence rather than a confidence impression.

NEXT ACTION

Open the latest metrics-pipeline transcript, mark the three decision moments, rehearse the corrected opening once, and run one comparable transfer prompt before changing any other part of the answer.